IOT ASSIGNMENT

Q1) You are building a smart irrigation system that must monitor soil moisture, control a water pump, and send data to a mobile application every 5 seconds. You have both an Arduino UNO and an ESP32. Which controller would you choose and why? Would your answer change if internet connectivity was not required?

Q2) A smart factory contains 500 sensors distributed across a 2 km area. Compare Wi-Fi, Bluetooth, LoRa, and MQTT-based communication. Which protocol or combination of protocols would you choose and why?

Q3) A temperature sensor suddenly starts showing impossible values (e.g., –120°C or 300°C). How you would determine whether the problem lies in the sensor, wiring, power supply, ADC, or software.

Answer 1: SMART IRRIGATION SYSTEM:
Arduino UNO v/s ESP32

Choice: ESP32

Reason:

- Built-in Wi-Fi and Bluetooth for IOT.
- Faster processor (240 MHz dual core) than Arduino UNO (16MHz).
- More RAM and Flash memory.
- Can easily send data to cloud every 5 seconds
- supports multitasking using FreeRTOS.

If internet is not required:

- Arduino UNO is sufficient if only sensor reading and pump control are needed.
- It is cheaper, consumes less power, and is easier for simple applications.

Answer 2.

Protocol	Range	Speed	Power	Suitable?
Wi-Fi	100m	High	High	Gateway communication
Bluetooth	10-100m	Medium	Low	Nearby devices
LoRa	2-15km	Low	very low	Large factory sensors
MQTT	Application protocol	depends on network	Low overhead	cloud communication

Best choice-
Sensors → LoRa
Gateway → Wi-Fi/Ethernet
Cloud communication → MQTT

Reason-
- LoRa covers entire 2km factory.
- MQTT efficiently transfers data to server
- Wi-Fi connects gateway to Internet.

Answer3:

1. Sensor:
   - replace with another sensor
   - compare readings
2. Wiring:
   - loose connections
   - Broken wire
   - wrong pin connection
3. Power Supply:
   - measures voltage using multimeter
   - ensure stable 3.3V or 5V.
4. ADC:
   - read another analog sensor.
   - check ADC calibration
5. Software:
   - verify pin number
   - verify conversion formula
   - Print raw values in Serial Monitor

Q4) An ESP32-based weather station must operate on battery power for six months without maintenance. What hardware and software modifications would you make to maximize battery life?

Q5) An automatic braking system in a vehicle requires communication with almost zero delay. Which communication method would you choose and why?

Answer 4.

HARDWARE.

- use Deep Sleep Mode.
- low-Power sensors
- Efficient voltage regulators
- Large Li-ion battery
- solar charging

SOFTWARE.

wake every few minutes
disable Wi-Fi except during upload.
reduce OLED usage
reduce CPU frequency.
upload data in batches.

Answer 5.

CAN Bus (Controller Area Network)
because

- very low latency
- highly reliable
- error detection
- used in automobiles
- works even under electrical noise

→ Wi-Fi or Bluetooth are unsuitable because of unpredictable delay.

Q6) You need to measure the average temperature of a classroom using only one sensor. Where would you place the sensor and why? Explain how poor placement could affect the accuracy of your measurements. Assume the classroom to be a square-base cuboid of side length “a” and height “b”.

Q7) Under what situations could an analog sensor perform better than a digital one?

Q8) Draw the complete architecture/block diagram of your IoT project, beginning from the sensor and ending at the user interface. Identify every possible point where a failure could occur and suggest one solution for each.

Answer 6:

Place sensor:

- near the center of classroom.
- Height ≈ 1.2-1.5m above floor.
- away from windows, fans, AC, heaters and sunlight.

For classroom dimensions:

Square base = $a \times a$

Height = b

Best position: $\left(\frac{a}{2}, \frac{a}{2}, \frac{b}{2}\right)$

Poor placement causes:

- hot readings near lights
- cold readings near AC.
- sunlight increases temperature
- Near door gives fluctuating values.

Answer 7:

Analog sensor performs better when:

- very fast response required
- continuous measurements needed.
- low-cost applications
- Simple circuits
- High resolution ADC available.

Example:
- Potentiometer
- LDR
- Gas sensor
- Soil Moisture sensor

Answer 8:

Transformer
↓
Sensors (DS18B20, MQ2)
↓
Edge Processing (ESP-32-S3)
↓
Gateway (Local IP)

Potential Issues:
1. sensor degradation from oil exposure.
2. EMI damaging ESP/ giving false reading

Solution:
1. use oil-resistant industrial sensors
2. use shielded wiring, proper grounding.

Q9) If an attacker gains access to your IoT device through the Wi-Fi network, what kinds of attacks could they perform? Suggest at least five methods to improve the security of the system (with applicable solutions, do not cite far-fetched ideas)

Q10) Your ESP32 must simultaneously interface with a DHT11 sensor, an ultrasonic sensor, an OLED display, a relay module, and a soil moisture sensor. What hardware or software conflicts could occur, and how would you solve them?

Q11) An ESP32 uploads sensor data to the cloud every minute. The internet connection fails for 12 hours. Design a strategy that ensures no important data is permanently lost.

Answer 9:

Possible Attacks:

- unauthorized control
- data theft
- Password stealing
- firmware modification
- denial of service (DoS)
- fake sensor data

Security improvements:

1) WPA2/WPA3 Wi-Fi
2) Strong Passwords
3) HTTPS/TLS encryption
4) Secure OTA updates
5) Disable unused ports
6) authentication tokens/API keys
7) regular firmware updates

Answer 10:

Problem — Conflicts:

Answer 10:

Problem	Solution
① Pin conflict	Assign different GPIOs
② Timing conflict	Use millis() instead of delay()
③ I2C conflict	unique addresses
④ Relay noise	use flyback diode
⑤ ADC interface	separate analog wiring
⑥ Power overload	External power supply

Answer 11:

① Read sensor every minute.

② Save readings in:

- SPIFFS
- LittleFS
- SD card

③ Retry Wi-Fi periodically.

④ After reconnect:

- upload oldest data first

⑤ Delete only after successful upload.

→ No data is lost.

Q12) Your IoT device must operate continuously for five years without human intervention. What changes would you make to both the hardware and software to improve long-term reliability?

Q13) Can an IoT system be considered "smart" if it does not use Artificial Intelligence? Justify your answer with suitable examples. What differentiates “smart” from “intelligent”?

Answer 12:

HARDWARE:

- industrial-grade sensors
- Quality PCB
- Surge protection
- Waterproof enclosure
- Watchdog circuit
- Good power supply

SOFTWARE:

- Watchdog timer
- Error logging
- Auto restart
- OTA updates
- Memory ~~too~~ leak prevention
- Retry mechanism

Answer 13:

Yes, A system can be smart without AI.

Example:

- Motion sensor turns ON lights.
- Temperature threshold starts fan.

→ This is rule-based automation.

AI becomes intelligent when it:

- learns patterns.
- predicts future events
- makes adaptive decisions

Difference:

Smart	Intelligent
rule based	learns automatically
fixed logic	AI/ML based
no learning	self-improving

Q14) Compare polling and interrupt-based programming in embedded systems. In which situations would interrupts significantly improve the performance of an IoT device?

Answer 14:

POLLING: The CPU repeatedly checks a condition in a loop. It's simply to implement but wastes CPU cycles and power when the event is infrequent, and can even miss very short events if polling rate isn't fast enough.

INTERRUPT-BASED: A hardware signal immediately triggers a dedicated interrupt service routine (ISR) the instant an event occurs, letting the CPU do other work. This is far more power- and time-efficient.

Interrupts significantly help in: battery-powered devices that need to sleep between rare events; fast, safety-critical events needing an instant response; and multitasking situations where the CPU shouldn't be stuck busy-waiting on one peripheral while networking or a dashboard update also needs attention.

Q15) Every engineering design involves trade-offs. Looking at your own IoT project, identify three design decisions where you sacrificed one aspect (such as cost, speed, power, accuracy, or complexity) to improve another. Explain whether you believe those trade-offs were justified.

Q16) Create a flowchart for an IoT system that stores sensor readings locally whenever Wi-Fi is unavailable and automatically uploads the stored data once the connection is restored. Mention all the programs/protocols/I-O used in the system.

Q17) Draw a flowchart showing how an ESP32 performs an OTA (Over-the-Air) firmware update, including error handling if the update fails.

Answer 15: design Trade-Offs

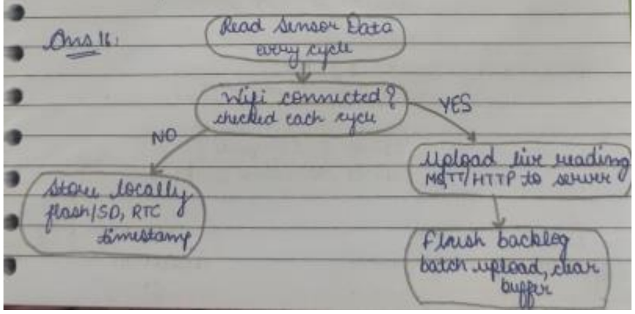
1. Cost v/s Accuracy: used MQ-2
cheaper but less accurate than DGA

2. Simplicity v/s features: used ESP32 local dashboard ~~for ThingSpeak~~
easy implementation but fewer visualization features.

3. Power v/s Continuous Monitoring:
ESP 32 keeps Wi-Fi ON
provides real-time monitoring but consumes more battery.

→ These trade-offs were justified because the project focused on low cost and ease of implementation.

Ans 16:

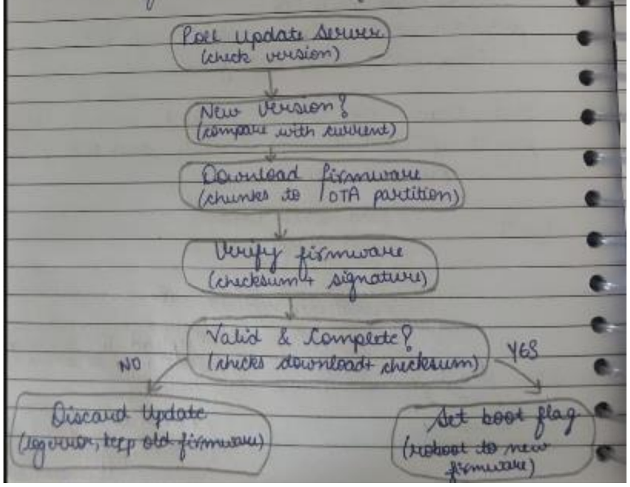


Programs/ Protocols/ I-O used:

- ESP 32
- DHT11 library
- Wi-Fi
- HTTP/ MQTT
- Thingspeak API
- SPIFFS/ SD card
- GPIO
- ADC
- I2C (OLED)

Answer 17:

OTA firmware update flowchart.



Error handling:

- verify firmware checksum.
- keep previous firmware for rollback.
- retry download if interrupted.
- log update failure.
- continue with old firmware if update fails